THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **Symmetry** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 4	A regular polygon with 4 sides has 4 lines of symmetry.
2	**(b)** 1	Letter A.
3	**(c)** F	It cannot be folded into two matching halves.
4	**(d)** 180	Angle $= 360^\circ \div$ order.
5	**(a)** infinitely many lines of symmetry	Symmetry property
6	**(c)** 3	Order = number of times it fits.
7	**(c)** 3	A regular polygon with 3 sides has 3 lines of symmetry.
8	**(a)** 2	Letter H.
9	**(d)** G	It cannot be folded into two matching halves.
10	**(a)** 120	Angle $= 360^\circ \div$ order.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A regular polygon has as many lines as sides (1); 4 lines (1).	1+1
12	Angle $= 360^\circ \div 4$ (1); 90° (1).	1+1
13	Fold along a vertical line (1); A, H, T (1).	1+1
14	A regular polygon has as many lines as sides (1); 6 lines (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Lines $= 4$ (1); Order $= 4$ (1); Smallest angle $= 90^\circ$ (1).	1+1+1
16	Imagine folding the letter (1); A: yes; H: yes; B: yes; C: yes (1); Check each fold (1).	1+1+1
17	A mirror image is at equal distance on the other side (1); The image is 2 units to the right of the mirror (1); Distance between point and image $= 2 \times 2 = 4$ units (1).	1+1+1
	OR: Lines $= 3$ (1); Order $= 3$ (1); Smallest angle $= 120^\circ$ (1).	1+1+1
18	Imagine folding the letter (1); H: yes; B: yes; C: yes; O: yes (1); Check each fold (1).	1+1+1
	OR: A mirror image is at equal distance on the other side (1); The image is 3 units to the right of the mirror (1); Distance between point and image $= 2 \times 3 = 6$ units (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Lines of symmetry $= 3$ (1); Order $= 3$ (1); Angle $= 360 \div 3 = 120^\circ$ (1); The polygon is a 3-sided regular polygon (2).	1+1+1+2
	OR: Check both folds (1); (a) H, O, X (1); (b) H, O, X, Z, N (1); Order 2 means it fits after a half turn (2).	1+1+1+2
20	Lines of symmetry $= 4$ (1); Order $= 4$ (1); Angle $= 360 \div 4 = 90^\circ$ (1); The polygon is a 4-sided regular polygon (2).	1+1+1+2
	OR: Check both folds (1); (a) O, X (1); (b) O, X, Z, N, S (1); Order 2 means it fits after a half turn (2).	1+1+1+2

Mistakes to avoid:

- Saying a rectangle has 4 lines of symmetry (diagonals are not lines of symmetry); a non-square rectangle has only 2.
- Giving a parallelogram 2 lines of symmetry; it has none, though it has rotational symmetry of order 2.
- Mixing up the order of rotation with the smallest angle: angle $= 360^\circ \div$ order.
- Saying a figure with order 1 has rotational symmetry; order 1 means only the trivial full turn.
- Drawing the mirror image at the wrong distance from the mirror line or on the same side of it.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **Symmetry** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 2	Symmetry property
2	**(c)** 4	Order = number of times it fits.
3	**(c)** 6	A regular polygon with 6 sides has 6 lines of symmetry.
4	**(c)** 1	Letter T.
5	**(d)** J	It cannot be folded into two matching halves.
6	**(b)** 90	Angle $= 360° \div$ order.
7	**(b)** 0	Symmetry property
8	**(d)** 5	Order = number of times it fits.
9	**(c)** 5	A regular polygon with 5 sides has 5 lines of symmetry.
10	**(c)** 1	Letter E.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Angle $= 360° \div 6$ (1); $60°$ (1).	1+1
12	Fold along a vertical line (1); H, T, X (1).	1+1
13	A regular polygon has as many lines as sides (1); 3 lines (1).	1+1
14	Angle $= 360° \div 3$ (1); $120°$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Lines $= 6$ (1); Order $= 6$ (1); Smallest angle $= 60°$ (1).	1+1+1
16	Imagine folding the letter (1); B: yes; C: yes; O: yes; S: no (1); Check each fold (1).	1+1+1
17	A mirror image is at equal distance on the other side (1); The image is 4 units to the right of the mirror (1); Distance between point and image $= 2 \times 4 = 8$ units (1).	1+1+1
	OR: Lines $= 5$ (1); Order $= 5$ (1); Smallest angle $= 72°$ (1).	1+1+1
18	Imagine folding the letter (1); C: yes; O: yes; S: no; X: yes (1); Check each fold (1).	1+1+1
	OR: A mirror image is at equal distance on the other side (1); The image is 5 units to the right of the mirror (1); Distance between point and image $= 2 \times 5 = 10$ units (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Lines of symmetry $= 5$ (1); Order $= 5$ (1); Angle $= 360 \div 5 = 72°$ (1); The polygon is a 5-sided regular polygon (2).	1+1+1+2
	OR: Check both folds (1); (a) X, I (1); (b) X, Z, N, S, I (1); Order 2 means it fits after a half turn (2).	1+1+1+2
20	Lines of symmetry $= 6$ (1); Order $= 6$ (1); Angle $= 360 \div 6 = 60°$ (1); The polygon is a 6-sided regular polygon (2).	1+1+1+2
	OR: Check both folds (1); (a) I (1); (b) Z, N, S, I (1); Order 2 means it fits after a half turn (2).	1+1+1+2

Mistakes to avoid:

- Saying a rectangle has 4 lines of symmetry (diagonals are not lines of symmetry); a non-square rectangle has only 2.
- Giving a parallelogram 2 lines of symmetry; it has none, though it has rotational symmetry of order 2.
- Mixing up the order of rotation with the smallest angle: angle $= 360° \div$ order.
- Saying a figure with order 1 has rotational symmetry; order 1 means only the trivial full turn.
- Drawing the mirror image at the wrong distance from the mirror line or on the same side of it.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **Symmetry** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** L	It cannot be folded into two matching halves.
2	**(d)** 60	Angle $= 360^\circ \div$ order.
3	**(a)** 1	Symmetry property
4	**(c)** 6	Order = number of times it fits.
5	**(b)** 8	A regular polygon with 8 sides has 8 lines of symmetry.
6	**(b)** 2	Letter X.
7	**(a)** P	It cannot be folded into two matching halves.
8	**(b)** 45	Angle $= 360^\circ \div$ order.
9	**(c)** reflection	Symmetry property
10	**(d)** 12	Order = number of times it fits.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Fold along a vertical line (1); T, X, M (1).	1+1
12	A regular polygon has as many lines as sides (1); 5 lines (1).	1+1
13	Angle $= 360^\circ \div 8$ (1); 45° (1).	1+1
14	Fold along a vertical line (1); T, X, M (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Lines $= 8$ (1); Order $= 8$ (1); Smallest angle $= 45^\circ$ (1).	1+1+1
16	Imagine folding the letter (1); O: yes; S: no; X: yes; Z: no (1); Check each fold (1).	1+1+1
17	A mirror image is at equal distance on the other side (1); The image is 6 units to the right of the mirror (1); Distance between point and image $= 2 \times 6 = 12$ units (1).	1+1+1
	OR: Lines $= 10$ (1); Order $= 10$ (1); Smallest angle $= 36^\circ$ (1).	1+1+1
18	Imagine folding the letter (1); S: no; X: yes; Z: no; A: yes (1); Check each fold (1).	1+1+1
	OR: A mirror image is at equal distance on the other side (1); The image is 7 units to the right of the mirror (1); Distance between point and image $= 2 \times 7 = 14$ units (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Lines of symmetry $= 8$ (1); Order $= 8$ (1); Angle $= 360 \div 8 = 45^\circ$ (1); The polygon is a 8-sided regular polygon (2).	1+1+1+2
	OR: Check both folds (1); (a) I (1); (b) N, S, I (1); Order 2 means it fits after a half turn (2).	1+1+1+2
20	Lines of symmetry $= 12$ (1); Order $= 12$ (1); Angle $= 360 \div 12 = 30^\circ$ (1); The polygon is a 12-sided regular polygon (2).	1+1+1+2
	OR: Check both folds (1); (a) I, H (1); (b) S, I, H (1); Order 2 means it fits after a half turn (2).	1+1+1+2

Mistakes to avoid:

- Saying a rectangle has 4 lines of symmetry (diagonals are not lines of symmetry); a non-square rectangle has only 2.
- Giving a parallelogram 2 lines of symmetry; it has none, though it has rotational symmetry of order 2.
- Mixing up the order of rotation with the smallest angle: angle $= 360^\circ \div$ order.
- Saying a figure with order 1 has rotational symmetry; order 1 means only the trivial full turn.
- Drawing the mirror image at the wrong distance from the mirror line or on the same side of it.

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